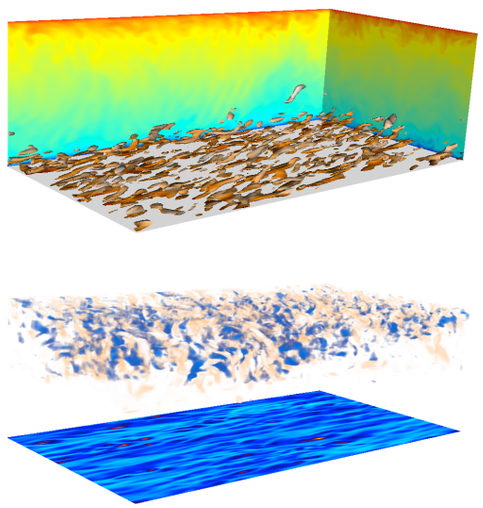
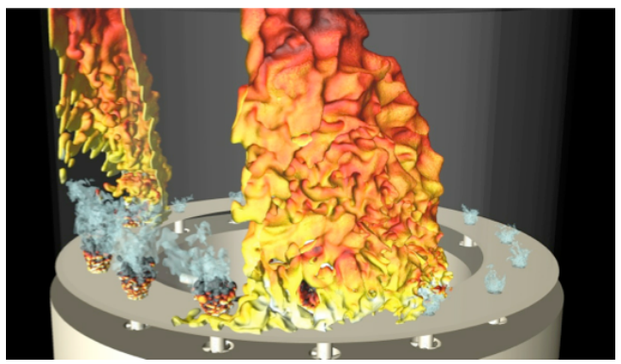
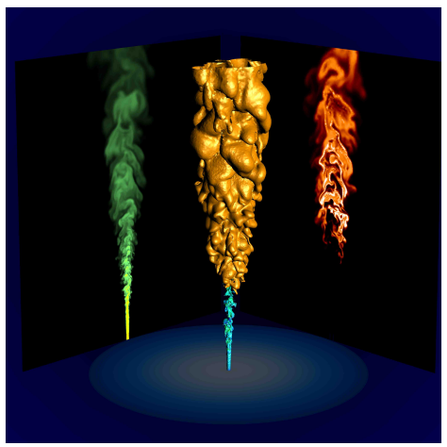
Computational Fluid Dynamics

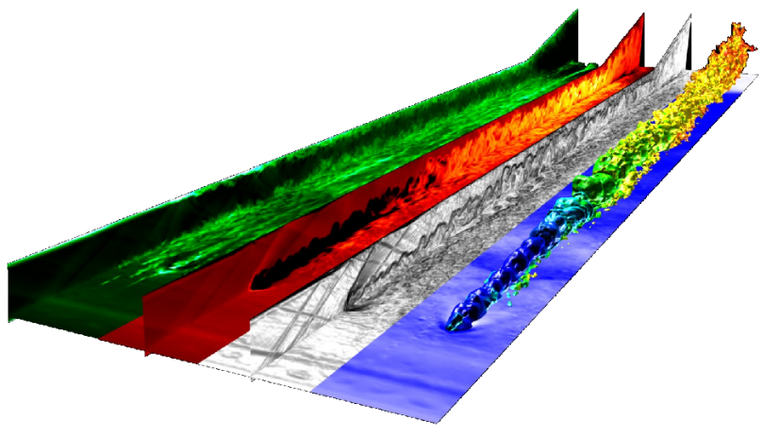
Final project

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Navier-Stokes equation in 2D

- Resolution of the full flow => incompressible Navier-Stokes equations

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0,$$

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} = -\frac{1}{\rho}\frac{\partial p}{\partial x} + \nu\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right)$$

$$\frac{\partial v}{\partial t} + u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} = -\frac{1}{\rho}\frac{\partial p}{\partial y} + \nu\left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2}\right)$$

Fractional step method

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0,$$

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} = -\frac{1}{\rho}\frac{\partial p}{\partial x} + \nu\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right)$$

$$\frac{\partial v}{\partial t} + u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} = -\frac{1}{\rho}\frac{\partial p}{\partial y} + \nu\left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2}\right)$$

step 1: transport

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} = 0$$

$$\frac{\partial v}{\partial t} + u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} = 0$$

hyperbolic

$$u^n \rightarrow u^*$$

$$v^n \rightarrow v^*$$

step 2: diffusion

$$\frac{\partial u}{\partial t} = \nu\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right)$$

$$\frac{\partial v}{\partial t} = \nu\left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2}\right)$$

parabolic

$$u^* \rightarrow u^{**}$$

$$v^* \rightarrow v^{**}$$

step 3: pressure

$$\frac{\partial u}{\partial t} = -\frac{1}{\rho}\frac{\partial p}{\partial x}$$

$$\frac{\partial v}{\partial t} = -\frac{1}{\rho}\frac{\partial p}{\partial y}$$

but you don't have P….

$$u^{**} \rightarrow u^{n+1}$$

$$v^{**} \rightarrow v^{n+1}$$

You can also mix steps 1 and 2 into a unique step

Pressure step

$$\frac{\partial u}{\partial t} = -\frac{1}{\rho}\frac{\partial p}{\partial x}$$
$$\frac{\partial v}{\partial t} = -\frac{1}{\rho}\frac{\partial p}{\partial y}$$

time discretisation

$$u^{n+1} = u^{**} - \frac{\Delta t}{\rho}\frac{\partial p}{\partial x}$$
$$v^{n+1} = v^{**} - \frac{\Delta t}{\rho}\frac{\partial p}{\partial y}$$

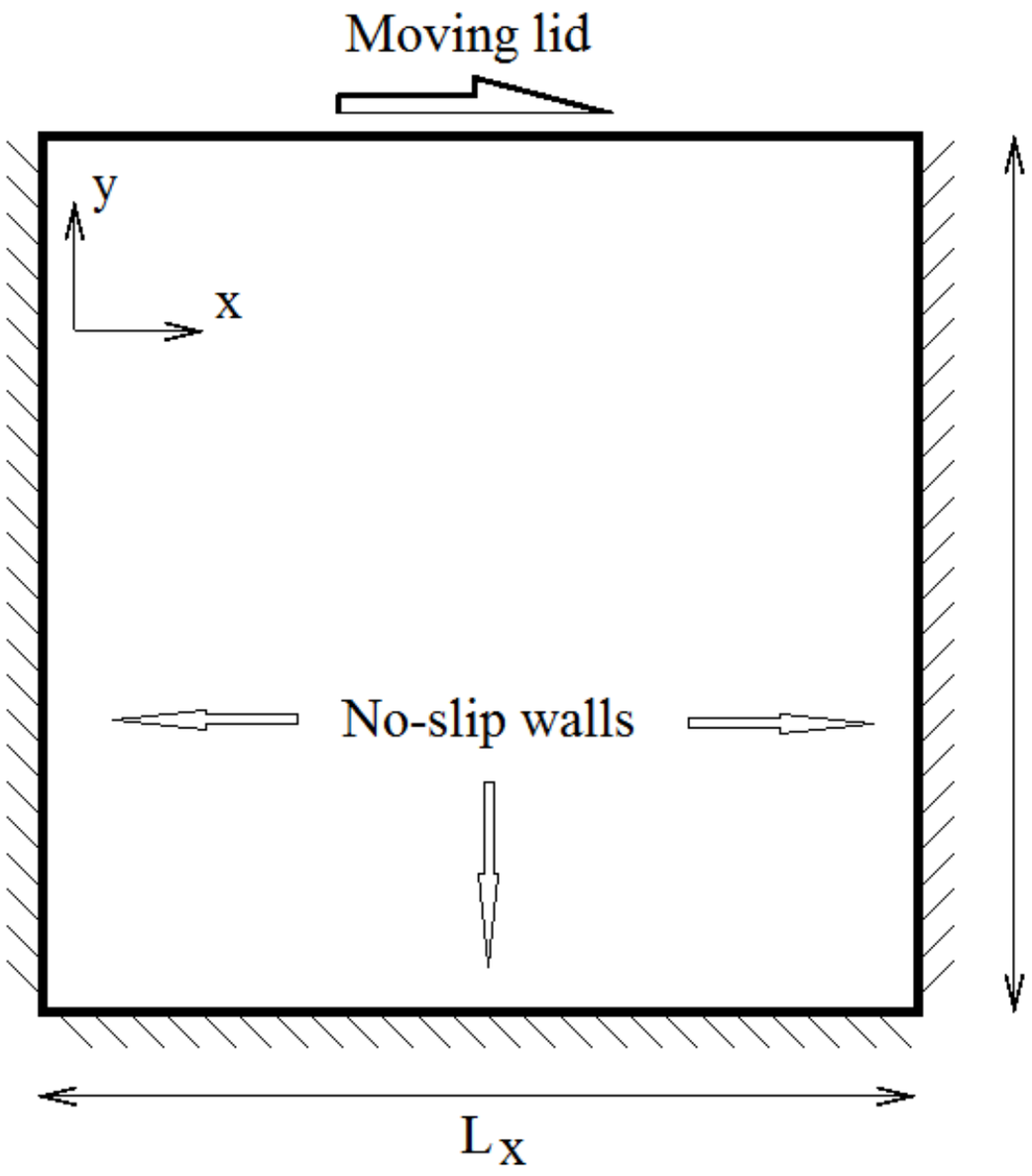
objective: divergence free update at n+1

$$\partial_x u^{n+1} + \partial_y v^{n+1} = 0$$

$$\partial_x^2 P + \partial_y^2 P = \frac{\rho}{\Delta t}(\partial_x u^{**} + \partial_y v^{**})$$

elliptic equation

The project: Lid-driven Cavity



Moving lid on top

$$u = U(t), \quad v = 0, \quad \frac{\partial P}{\partial n} = 0$$

No-slip walls

$$u = 0, \quad v = 0, \quad \frac{\partial P}{\partial n} = 0$$

Pressure equation with Neumann condition only

Pressure equation

$$\partial_x^2 P + \partial_y^2 P = \frac{\rho}{\Delta t}(\partial_x u^{**} + \partial_y v^{**}) \qquad \frac{\partial P}{\partial n} = 0$$

Existence of a solution but no unicity

If P is a solution P+C is also a solution

One of the possible solutions: impose a zero mean

At the end of each iteration of the pressure solver, subtract mean(P) to P

Evaluation

Presentation + code to be sent before december, 1st

Defense:

- 30 min
- 10 min presentation, 20 min questions
- 2 time slots:
  - Wednesday, December, 2th from 2pm to 5pm
  - Thursday, December, 3rd from 4.30pm to 6pm